RAMOA: Risk-Adaptive Multi-Agent Orchestration for Autonomous Security Operations

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1. Project Background and Motivation

Modern Security Operations Centers process enormous volumes of telemetry including process logs, network flows, authentication events and file activities. Traditional SOC systems rely on:

- static rule-based detection
- fixed correlation pipelines
- manual triage
- uniform execution of all detection modules

These approaches suffer from:

- alert fatigue
- high false-positive rates
- poor resource allocation
- lack of adaptive prioritization
- excessive computational overhead

Recent trends introduce AI-driven automation and multi-agent systems. However, most systems either:

- run all detection agents on all events, or
- rely on opaque AI decision-making without deterministic control

Both approaches lack structured, cost-aware and trust-aware orchestration.

RAMOA addresses this gap by introducing a **risk-adaptive orchestration algorithm** that dynamically selects, sequences and evaluates analytical agents based on expected utility, trust scoring and structural anomaly context.

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2. Core Problem Statement

How can a Security Operations Center dynamically determine:

- which analytical modules to execute
- how deeply to analyze an event
- how much to trust each analytical component
- when to escalate analysis
- how to reduce unnecessary computation

without relying on static routing rules or opaque AI-only reasoning?

3. Core Contribution

RAMOA introduces a **Risk-Adaptive Multi-Agent Orchestration Algorithm** that:

1. Computes preliminary risk for each event.
2. Maintains persistent trust profiles for analytical agents.
3. Calculates expected utility for agent invocation.
4. Selectively invokes only high-utility agents.
5. Aggregates outputs using confidence-weighted fusion.
6. Updates agent trust scores through feedback.
7. Supports progressive analytical deepening.

The system optimizes three competing factors:

- detection accuracy
- computational cost
- response latency

4. High-Level System Architecture

```
Security Telemetry
        ↓
Event Normalization Layer
        ↓
Graph Construction Layer
        ↓
Graph Metrics Engine
        ↓
Risk Estimation Module
        ↓
RAMOA Orchestrator (Core Algorithm)
        ↓
Selective Agent Invocation
        ↓
Confidence Fusion Layer
        ↓
Incident Assessment & Reporting
        ↓
Trust Update & Learning Loop
```

5. Detailed Component Architecture

5.1 Security Telemetry Ingestion Layer

Collects:

- Process execution logs
- Command-line activity
- Network connection metadata
- Authentication events
- File hash activity

Logs are normalized into a canonical internal event schema:

```
Event {
    event_id
    timestamp
    host_id
    user_id
    process_id
    parent_process_id
    process_name
    command_line
    destination_ip
    file_hash
    event_type
}
```

5.2 Graph Construction Layer

Security events are transformed into a dynamic temporal graph.

Node Types:

- Process
- User
- File
- IP address
- Host

Edge Types:

- spawned
- executed_by
- connected_to
- wrote_file
- authenticated_to

Example structure:

User → spawned → PowerShell
PowerShell → connected_to → External IP
PowerShell → wrote_file → DLL

The graph is maintained within a sliding time window.

5.3 Graph Metrics Engine

Graph-based anomaly features are computed using NetworkX.

Metrics include:

- Degree centrality
- Betweenness centrality
- Rare edge frequency
- Subgraph density spikes

Graph Anomaly Score:

$$GraphAnomalyScore = \alpha \times DegreeDeviation$$

$$\beta \times BetweennessDeviation$$

$$\gamma \times RarityScore$$

$$\delta \times DensitySpike$$

This score feeds into risk estimation.

5.4 Risk Estimation Module

Combines:

- heuristic risk signals
- anomaly detection outputs
- graph anomaly score

Final estimated risk:

$$EstimatedRisk = w1 \times HeuristicRisk$$

$$w2 \times GraphAnomalyScore$$

This module does not produce final verdicts.
It only guides orchestration.

5.5 Analytical Agent Layer

Contains modular analytical agents such as:

- Malware heuristic agent

- Network anomaly agent
- Process chain agent
- Threat intelligence lookup agent

Each agent outputs:

```
AgentResult {
    verdict
    confidence_score
    evidence
}
```

Agents are independent and implementation-agnostic.

5.6 Agent Profile & Trust Store

Each agent has a persistent profile:

```
AgentProfile {
    trust_score
    execution_cost
    latency_metric
    false_positive_rate
}
```

Trust score is dynamically updated.

5.7 RAMOA Orchestrator (Core Algorithm)

For each event, compute:

$$ExpectedUtility(Ai) = (EventRisk \times AgentTrust) - ExecutionCost - LatencyPenalty$$

Invoke agent Ai if:

ExpectedUtility ≥ threshold

This ensures:

- cost-aware invocation
- trust-aware selection
- dynamic control
- resource optimization

5.8 Confidence Fusion Layer

Agent outputs are aggregated using confidence-weighted fusion:

$$FinalConfidence = \Sigma(confidence_i \times trust_score_i)/\Sigma(trust_score_i)$$

Supports progressive deepening:

If confidence exceeds escalation threshold → invoke deeper agents
Else → stop analysis

5.9 Learning & Adaptation Loop

After validation:

- increase trust for correct predictions
- penalize false positives
- penalize missed detections

System becomes self-optimizing without retraining ML models.

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6. Evaluation Plan

Experiments will demonstrate:

1. Static invocation vs RAMOA selective invocation
2. RAMOA with vs without graph metrics
3. Trust adaptation over time

Metrics:

- Precision
- Recall
- F1-score
- False positive rate
- Mean time to decision
- Average agents invoked per event

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7. Technical Stack

- Python
- NetworkX for graph modeling
- Scikit-learn for lightweight anomaly detection
- SQLite / JSON for persistence
- Optional LLM for report generation
- LangGraph optional for workflow control

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8. Project Phases

Phase 1 – Core RAMOA engine
Phase 2 – Graph integration

Phase 3 – Agents + evaluation
Phase 4 – Paper + patent preparation

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9. Patent Direction

Core patentable elements:

- Utility-based selective agent invocation
- Adaptive trust-based orchestration
- Confidence-weighted fusion
- Progressive analytical deepening
- Graph-informed risk adaptation

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10. Real-World Significance

RAMOA provides:

- reduced SOC workload
- improved resource allocation
- adaptive reliability
- explainable automated decision-making

It aims to bridge academic multi-agent research with operational SOC realities.